# COHOMOLOGICAL TOOLS FOR UNDERSTANDING MOD $p$ REPRESENTATIONS OF $p$-ADIC LIE GROUPS

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## 1. Introduction

For over 50 years, the Langlands program has been a driving force of research in number theory, deeply intertwining numerous parts of the field including Galois groups of number fields, modular forms, elliptic curves, automorphic forms, and the representation theory of reductive groups over local fields. While all of these topics are rich and deeply studied on their own, the connections which the Langlands program conjectures have become some of the most exciting recent developments in number theory. For example, one special case of the Langlands program is the modularity theorem for elliptic curves, which was famously used to prove Fermat's Last Theorem.

In recent years, the mod $p$ local Langlands program has inspired much progress in understanding the mod $p$ representation theory of $p$-adic reductive groups (such as $\mathrm{GL}_n(\mathbb{Q}_p)$), and conjectures a connection between it and the mod $p$ representation theory of absolute Galois groups of $p$-adic fields. Many facts which have long been known in the classical case where representations are over $\mathbb{C}$ are no longer true when one's representations are over a field of characteristic $p$, and finding analogues of many of these facts has naturally led many authors to consider cohomological tools and derived categories (e.g., see Subsection 3.2 below). Many questions which are easy to state remain open, making this an exciting area for future research.

During my PhD, my research has focused on answering cohomological questions which naturally come up in the mod $p$ representation theory of $p$-adic reductive groups, and developing the tools required to do so. In this statement, I will motivate and present the main results of two preprints.

In the first preprint, described in Section 2, I take a construction of a $p$-valuation on a pro-$p$ Iwahori subgroup of a $p$-adic reductive group, which was initially performed in [Laz65] and later generalized in [LS24], and I push it to proving a result which is nearly the best one could hope for. In doing so, I also connect the Moy-Prasad theory of reductive groups over $p$-adic fields deeply to Lazard theory, the kernel of which was first noted in [DLH25, Section 3], but which is now greatly generalized. The primary results I have proven are Theorems 2.2 and 2.4 below.

In the second preprint, which is joint work with Hargun Bhatia, Yashi Jain, and Ben Savoie and which is described in Section 3, our main goal was to explicitly compute a family of cohomological mod $p$ Satake transforms constructed in an abstract way in [Hey23] (which had as special cases constructions due to [Her11b] and [HV15]). In particular, we find explicit formulas for Heyer's Satake transforms, simplify them greatly when working with the reductive group $\mathrm{GL}_n(F)$ for $F/\mathbb{Q}_p$ a finite extension, and then perform very precise computations when working in the case of $\mathrm{GL}_2(F)$. All of these results are presented sequentially in Subsection 3.5.

1.1. **Acknowledgments.** I am deeply grateful for the years of guidance, encouragement, and support provided by my advisor Claus Sorensen during my PhD. Without this, I certainly would not have been in a position to contribute any part of the results described below.

## 2. $p$-valuations and the Moy-Prasad filtration

Let $k$ be a field of characteristic $p$. Let $F/\mathbb{Q}_p$ be a finite extension with ring of integers $\mathcal{O}_F$ and absolute ramification index $e$. We endow $F$ with a valuation $v$, normalized so that $v(p) = 1$.

2.1. **History of $p$-valuations.** The notion of a *$p$-valuation* on a pro-$p$ group $G$, and the additional properties that $G$ satisfies when the $p$-valuation is *saturated*, is an incredibly useful tool for understanding $p$-adic Lie groups. In his 1965 paper [Laz65], Lazard is able to use this tool to prove several significant theorems, the most relevant for our purposes involving the classification of $p$-adic Lie groups (see [Sch11, Theorem 27.1, Theorem 29.8]), and about algebraic properties of the

completed group ring $k[\![G]\!]$ (see [Sch11, Theorem 33.4] for the integral analogue). A more modern treatment of these same ideas, which is also in English, is presented in Part B of the recent text [Sch11] (which is why we are citing this rather than the original locations in [Laz65] where these facts appear).

In recent decades, the study of saturated pro-$p$ groups has become especially relevant, especially in representation theory. The basis for much of this work originates from [Laz65, 3.2.7], where Lazard proves that the standard pro-$p$ Iwahori, i.e., a particular maximal pro-$p$ subgroup $I$ of $\mathrm{GL}_n(\mathcal{O}_F)$ and $\mathrm{SL}_n(\mathcal{O}_F)$, is saturable as long as $p-1 > en$. Lazard also notes that this bound on $p$ is sharp: there are examples with $p-1 = en$ for which $I$ admits no $p$-valuation, let alone a saturated one.

In [Tot99, Theorem 12.1], Totaro takes Lazard's argument further and shows that every pro-$p$ subgroup of a semisimple algebraic group defined over $F$ admits a $p$-valuation when $p >> 0$ (with an explicit lower bound for $p$), however he does not prove that his $p$-valuation is saturated on a full pro-$p$ Iwahori subgroup. Having a saturated $p$-valuation is much more useful than merely having a $p$-valuation; for example, naturally associated to a saturated pro-$p$ group is a Lie $\mathbb{Z}_p$-algebra (see [Sch11, Section 31]). More geometrically, [LS24, Proposition 4.1] shows that there is a natural rigid analytic group over $\mathbb{Q}_p$ associated to any saturated pro-$p$ group, which is latter promoted to a dagger group over $\mathbb{Q}_p$ in [LSS26, Proposition 5.4]. The issue of saturation is addressed in a slightly different context in [LS24, Proposition 3.5]. Concretely, they show the following.

**Theorem 2.1** (Lahiri-Sorensen)**.** *Let $\mathbf{G}$ be a split connected reductive $F$-group, $h$ its Coxeter number, and $I \subseteq \mathbf{G}(F)$ a pro-$p$ Iwahori subgroup. If $p-1 > eh$, then there exists a saturated $p$-valuation $\omega$ on $I$.*

While this addresses the saturation question, it requires that the group $\mathbf{G}$ be split. To prove this, they construct a particular $p$-valuation, prove that its restriction to each term in the Iwahori factorization

$$\prod_{\alpha\in\Phi^-} U_{\alpha,0+} \times T_{0+} \times \prod_{\alpha\in\Phi^+} U_{\alpha,0} \xrightarrow{\sim} I \tag{2.1}$$

is saturated, and prove that this is sufficient to conclude the saturation on $I$. It is noted in [DLH25, Remark 3.5] that this $p$-valuation arises from a particular Moy-Prasad filtration, which we discuss in more depth below in Subsection 2.3. In [Per26], I have generalized Theorem 2.1 in a few ways; see Theorem 2.2 below.

2.2. **History of Moy-Prasad filtrations.** Let $\mathbf{G}$ be a connected reductive group defined over our $p$-adic field $F$ with $G = \mathbf{G}(F)$, $\mathbf{S} \leq \mathbf{G}$ a maximal $F$-split torus, $\Phi = \Phi(\mathbf{G}, \mathbf{S})$ its root system. (For example, one could take $G = \mathrm{GL}_n(F)$ and $S = \mathbf{S}(F)$ its subgroup of diagonal matrices.) Let $x \in \mathcal{A}(\mathbf{S})$ be any fixed point in the apartment associated to $\mathbf{S}$ in the Bruhat-Tits building $\mathcal{B}(\mathbf{G})$. Then associated to any concave function $f : \Phi \cup \{0\} \to \mathbb{R}$ is a compact open subgroup $G_{x,f} \leq G$. (For definitions of all of these terms, see the recent text [KP23].) These were originally considered by Bruhat and Tits in the case when $f(0) = 0$ in their papers [BT72] and [BT84], and were later generalized beyond this requirement by Moy and Prasad in [MP94] and [MP96]. An important special case which is allowed once we remove the requirement that $f(0) = 0$ is when $f \geq 0$ is constant, i.e., there is some $r \geq 0$ such that $f = f_r \equiv r$ always outputs $r$. In this case, we write $G_{x,r} \coloneqq G_{x,f_r}$, and the resulting filtration on $G_{x,0}$ is called the *Moy-Prasad filtration*, with the $G_{x,r}$ being called the *Moy-Prasad filtration groups*. The construction of the Moy-Prasad filtration is later slightly altered by Yu in [Yu15], after which more desirable properties can be proven when the maximal torus $\mathbf{T}$ of $\mathbf{G}$ is not *weakly induced* (in particular, when it does not even split over a tamely ramified extension).

The Moy-Prasad filtration associated to a point $x \in \mathcal{B}(\mathbf{G})$ has become an incredibly important tool for the study of the representation theory of $G$, especially in the case when one is considering complex representations. For example, in [MP96], Moy and Prasad introduce the notion of the

*depth* of a representation, defined using Moy-Prasad filtrations. Later, [Adl98] and [Yu01] critically use this notion to construct many supercuspidal representations of $G$, which was shown in [Fin18] to produce all supercuspidal representations, so long as the group splits over a tame extension and $p$ does not divide the order of the Weyl group.

2.3. **Connections between $p$-valuations and Moy-Prasad filtrations.** We assume for technical reasons in this subsection that $p > 2$ is an odd prime. We take the same setup as in the previous subsection.

To fix notation, for $r \geq 0$ and any $x \in \mathcal{B}(\mathbf{G})$, write $G_{x,r+} = \bigcup_{s>r} G_{x,s}$ and define

$$\omega_x(g) \coloneqq \inf\{r > 0 \,|\, g \in G_{x,r}\}$$

for $g \in G_{x,0+}$. As noted earlier, [DLH25, Remark 3.5] points out that the $p$-valuation $\omega$ defined on the pro-$p$ Iwahori subgroup $I$ studied by Lahiri and Sorensen in Theorem 2.1 is actually equal to $\omega_x$ for a particular $x \in \mathcal{B}(\mathbf{G})$ (notably, the barycenter of the dominant base alcove), where $G_{x,0+} = I$ is exactly the pro-$p$ Iwahori subgroup studied in [LS24].

This poses an immediate natural question: do other points $x \in \mathcal{B}(\mathbf{G})$ give us saturated $p$-valuations on $G_{x,0+}$? Unfortunately, one can easily find counterexamples to this question; a requirement in the definition of $\omega_x$ being a $p$-valuation on $G_{x,0+}$ would need that $\omega_x(g) > \frac{1}{p-1}$ for all $g \in G_{x,0+}$, which forces $G_{x,0+} = G_{x,\frac{1}{p-1}+}$ (which is false for easy examples). However, if one is willing to restrict their view to the smaller group $G_{x,\frac{1}{p-1}+}$, then I prove in [Per26] that the answer is 'yes'. In fact, it turns out that the condition $p - 1 > eh$ present in Theorem 2.1 is ultimately there to ensure that $G_{x,0+} = G_{x,\frac{1}{p-1}+}$ for $x$ the barycenter of the dominant base alcove, allowing us to drop this restriction. Moreover, I am able to replace the split requirement with an assumption on the torus, and prove the following result:

**Theorem 2.2** (P)**.** *Let $\mathbf{G}$ be a connected reductive $F$-group, and assume that $\mathbf{T}$ is weakly induced. Then for any $x \in \mathcal{B}(\mathbf{G})$, $\omega_x$ is a saturated $p$-valuation on $G_{x,\frac{1}{p-1}+}$.*

We note that $\mathbf{T}$ is weakly induced if $\mathbf{G}$ is split, making this a strict generalization of Lahiri and Sorensen's Theorem 2.1. (This condition on $\mathbf{T}$ is also satisfied if $\mathbf{G}$ is quasisplit, semisimple, and either adjoint or simply connected. It also holds if $\mathbf{G}$ splits over a tame extension.)

If $\mathbf{G}$ is split, then the proof of this theorem proceeds very similarly to that of Lahiri-Sorensen's Theorem 2.1, except that one has to put slightly more effort into confirming that $\omega_x$ restricts to saturated $p$-valuations on each of the root groups $U_{\alpha,x,\frac{1}{p-1}+}$ and the torus $T_{\frac{1}{p-1}+}$ for $T = \mathbf{T}(F)$ (compare to the Iwahori factorization 2.1).

If $\mathbf{G}$ is quasisplit, these root groups break into two types, labeled R1 and R2 (see [KP23, 2.6.8] for details). The root groups of type R1 look like those in split groups and are similarly easy to understand. However, the root groups of type R2 are more challenging to understand; one of my main contributions in the case when $\mathbf{G}$ is quasisplit is proving that $\omega_x$ always restricts to a saturated $p$-valuation on these root groups of type R2.

If $\mathbf{G}$ is not even quasisplit, then by a result of Steinberg [Ste65], it becomes quasisplit after base changing to $\mathbf{G}_E$ along a finite unramified extension $E/F$. By [KP23, Proposition 9.8.3], we have for any $x \in \mathcal{B}(\mathbf{G}) = \mathcal{B}(\mathbf{G}_E)^{\mathrm{Gal}(E/F)}$ and any $r \geq 0$ that

$$G_{x,r} = (\mathbf{G}_E(E)_{x,r})^{\mathrm{Gal}(E/F)}.$$

In order to prove Theorem 2.2 in this case, I had to introduce a new tool for detecting when $p$-valuations are saturated in [Per26]:

**Lemma 2.3** (P)**.** *Let $(H, \omega)$ be a saturated group, and suppose that $\Theta$ is a group acting on $H$ via continuous automorphisms. Then $\omega$ restricts to a saturated $p$-valuation on the fixed points of this*

*action*

$$H^{\Theta} := \{h \in H \mid \theta(h) = h \text{ for all } \theta \in \Theta\}.$$

We also note for posterity that one could abstractly prove that there exists some saturated $p$-valuation on $G_{x,0+}$ if $p >> 0$ using a combination of Lahiri and Sorensen's Theorem 2.1 and [GSK09, Theorem A], however this has several drawbacks: it is not especially conducive to performing computations (as we do below for the Lie algebra), it provides no information about the case when $p$ is not quite large, and it would not provide us with information about the Moy-Prasad filtrations themselves.

2.4. **Lie algebra computations.** Via Lazard theory [Laz65], there is a Lie $\mathbb{F}_p[P]$-algebra $\operatorname{gr} G$ associated to any $p$-valued pro-$p$ group $G$ which encodes much of the information of the group. (Here, $P$ is merely viewed as a formal variable, making $\mathbb{F}_p[P]$ just a polynomial algebra.) One can similarly produce a Lazard Lie $\mathbb{F}_p$-algebra $\operatorname{gr} G \otimes_{\mathbb{F}_p[P]} \mathbb{F}_p = \operatorname{gr} G/P(\operatorname{gr} G)$. Two recent applications of Lahiri and Sorensen's Theorem 2.1 have relied deeply on explicit descriptions of $\operatorname{gr} G$ and $\operatorname{gr} G/P(\operatorname{gr} G)$ when $G$ is a pro-$p$ Iwahori, notably [DLH25] and [ACH$^+$26]. Both of these papers rely on similar intense ad-hoc computations to produce expressions for these Lie algebras, and have very similar outcomes. One point which has become clear from these two applications is that it is of great importance that we better understand these Lie algebras in order to apply Lazard theory to the mod $p$ representation theory of $p$-adic groups, and that applications of the new Theorem 2.2 will likely require this information.

By working more conceptually, I have been able to avoid the heavy computations present in [DLH25] and [ACH$^+$26]. In particular, I have been able to reinterpret the Moy-Prasad isomorphism as an isomorphism of Lie $\mathbb{F}_p[P]$-algebras. The kernel of this idea is present in the proof of [DLH25, Lemma 3.7], which still relies on a manual computation (that becomes completely untenable once one is no longer working in the split case).

Let now $p > 2$, $\mathbf{G}$ be a connected reductive $F$-group with maximal torus $\mathbf{T}$, and $\mathcal{B}(\mathbf{G})$ be its Bruhat-Tits building. Write $G = \mathbf{G}(F)$. Fix any $x \in \mathcal{B}(\mathbf{G})$, associated to which is a filtered Lie $\mathcal{O}_F$-algebra $\mathfrak{g}_{x,0}$ (defined using integral models of $\mathbf{G}$). We then have the following description of the Lie $\mathbb{F}_p[P]$-algebra of $G_{x,\frac{1}{p-1}+}$:

**Theorem 2.4** (P)**.** *Assume that* $\mathbf{T}$ *is weakly induced. The Moy-Prasad isomorphism*

$$\mathrm{MP}_x : \operatorname{gr} G_{x,\frac{1}{p-1}+} = \bigoplus_{r>\frac{1}{p-1}} G_{x,r}/G_{x,r+} \xrightarrow{\sim} \bigoplus_{r>\frac{1}{p-1}} \mathfrak{g}_{x,r}/\mathfrak{g}_{x,r+} = \operatorname{gr} \mathfrak{g}_{x,\frac{1}{p-1}+},$$

*a priori only known to be an isomorphism of* $\mathbb{F}_p$*-vector spaces (see* [KP23, Section 13.5] *for details), is in fact an isomorphism of Lie* $\mathbb{F}_p[P]$*-algebras.*

Here, we are viewing the Lie $\mathcal{O}_F$-algebra $\mathfrak{g}_{x,\frac{1}{p-1}+}$ as first being a module over $\mathbb{Z}_p$, and then using the natural isomorphism $\operatorname{gr} \mathbb{Z}_p \cong \mathbb{F}_p[P]$ sending $p \mapsto P$ to view $\operatorname{gr} \mathfrak{g}_{x,\frac{1}{p-1}+}$ as a Lie algebra over $\mathbb{F}_p[P]$.

2.5. **Cohomological applications and future directions.** Many recent applications of the $p$-valuation machinery to mod $p$ representation theory arise from [Sor21]. Given a $p$-valued pro-$p$ group $G$, let $\mathfrak{g}$ be its Lazard Lie algebra, which is finite dimensional if $G$ is of finite rank (see [Sch11, Section 25] for details; this is just $\mathfrak{g} = \operatorname{gr} G/P(\operatorname{gr} G)$). Then [Sor21, Theorem 5.5] constructs a multiplicative spectral sequence

$$E_1^{s,t} = H^{s,t}(\mathfrak{g}, k) \implies H^{s+t}(G, k), \tag{2.2}$$

where the terms on the $E_1$ page consist of graded Lie algebra cohomology, and which is converging to the continuous group cohomology of $G$. (The result in [Sor21] applies to more general coefficients; here, we only discuss the trivial one-dimensional representation $k$ for simplicity.)

This spectral sequence 2.2 was recently used in [DLH25] to fully compute the cohomology ring $H^*(\mathbf{G}(\mathcal{O}_F), k)$ when $k = k_F$ is the residue field, under the assumptions that $F/\mathbb{Q}_p$ is unramified, $p > h+1$, and $\mathbf{G}$ is a split reductive $F$-group. This relied on a reference to [LS24], forcing the split requirement. A good test case to see how much the techniques in [DLH25] can be pushed beyond the split case would be to attempt to compute the full cohomology ring of $\mathrm{SU}_3(\mathcal{O}_F)$ with the same hypotheses on $F$ and $p$, which is a small quasisplit (but non-split) group. (See [KP23, 2.7, 3.2] for notation.)

Another direction of recent research has been using Lazard theory for pro-$p$ Iwahori subgroups to obtain bounds on the Gelfand-Kirillov dimension of smooth admissible representations of $\mathbf{G}(F)$ when $\mathbf{G}$ is split, $F/\mathbb{Q}_p$ is unramified, and $p > h+1$. This was first done for $\mathbf{G} = \mathrm{GL}_2$ in [BHH$^+$23], but when [ACH$^+$26] attempted to generalize the argument to all split groups using a reference to [LS24], they note that the methods do not apply if the rank of the root system is greater than 1, meaning that they are not directly useful in the split case beyond $\mathrm{GL}_2$ (or $\mathrm{SL}_2$), already studied in [BHH$^+$23]. However, the rank of the root system of $\mathrm{SU}_3$ is also 1, meaning that the arguments of [BHH$^+$23] might apply to this case.

Finally, another natural direction would be to attempt to remove hypotheses on Theorems 2.2 and 2.4. That is, it would be appealing to allow $p = 2$ and any connected reductive group, dropping our condition on the torus. If $p = 2$, the definition of the Moy-Prasad filtration is more complicated, and a couple of arguments in the proof of Theorem 2.2 break down for technical reasons, however there do not seem to be any major theoretical problems for allowing $p = 2$ in that theorem. By contrast, whether we can drop our condition on the torus is much less clear. In the general case, the construction of the appropriate filtration on the torus (notably, the minimal congruent filtration of [Yu15]) is significantly more involved, and includes taking Néron models. New tools would almost certainly have to be developed to study this, making it quite an interesting direction to move in.

## 3. Higher mod $p$ Satake transforms

Given any $p$-adic Lie group $G$ and any field $k$, we write $\mathrm{Rep}_k(G)$ for the category of smooth representations of $G$ on $k$-vector spaces. We then define $D(G) \coloneqq D(\mathrm{Rep}_k(G))$ to be its derived category of unbounded cochain complexes.

Let now $k$ be a field of characteristic $p$, and $F/\mathbb{Q}_p$ a finite extension with ring of integers $\mathcal{O}_F$.

3.1. **History of Satake transforms.** The Satake transform has been a fundamental tool for studying the complex representation theory of $p$-adic reductive groups since its introduction by Satake in [Sat63], relating the spherical Hecke algebra of the reductive group to a Hecke algebra for its maximal torus. As a key example, let $G = \mathrm{GL}_n(F)$, $T$ its subgroup of diagonal matrices, and write $K = \mathrm{GL}_n(\mathcal{O}_F)$, which is a maximal compact subgroup of $G$. Then the spherical Hecke algebra

$$\mathcal{H}(G, K) = C_c^\infty(K\backslash G/K, \mathbb{C})$$

is related to the analogous toral Hecke algebra by the classical Satake isomorphism

$$S^{\mathrm{cl}} : \mathcal{H}(G, K) \xrightarrow{\sim} \mathcal{H}(T, T \cap K)^W$$

(where we have taken Weyl invariants in the toral Hecke algebra). This allows one to better understand the complex representations of $G$ via the torus $T$, which is a much easier group to work with.

When considering representations over a field $k$ of characteristic $p$, one can similarly define the mod $p$ Hecke algebras

$$\mathcal{H}_G(k) = C_c^\infty(K\backslash G/K, k) \qquad \text{and} \qquad \mathcal{H}_T(k) = C_c^\infty(T/T \cap K, k)$$

(now dropping $K$ from our notation), and there is an injective Satake transform

$$\mathcal{S} : \mathcal{H}_G(k) \hookrightarrow \mathcal{H}_T(k)$$

constructed first by Herzig in [Her11b]. In fact, one can define Hecke algebras $\mathcal{H}_G(V)$ and $\mathcal{H}_T(V)$ for any Serre weight $V$, i.e., any smooth irreducible representation of $K$, and in this context the Satake transform Herzig constructs is an injective homomorphism $\mathcal{S} : \mathcal{H}_G(V) \hookrightarrow \mathcal{H}_T(V)$; here, Herzig only needs to assume that that $G = \mathbf{G}(F)$ for $\mathbf{G}$ a connected unramified reductive $F$-group and $K$ a hyperspecial subgroup. This later gets extended to all connected reductive $F$-groups (where $T$ is now replaced by any Levi subgroup) by Henniart-Vignéras in [HV15].

After its introduction, the mod $p$ Satake transform was rapidly applied to solving problems in the mod $p$ representation theory of $p$-adic reductive groups. For example, Herzig uses it critically to classify the mod $p$ irreducible admissible representations of $\mathrm{GL}_n(F)$ in terms of supersingular representations in [Her11a]. This work was later extended to split reductive $F$-groups in [Abe13], and later extended to all connected reductive $F$-groups in [AHHV17].

3.2. **Connection to derived categories.** Let $G$ be a $p$-adic Lie group, and let $K$ be a compact open subgroup. Given any complex representation $V$ of $G$, there is a natural action of the Hecke algebra $\mathcal{H}(G, K)$ on the invariants $V^K$. It is then a classical fact that the functor

$$(-)^K : \mathrm{Rep}_{\mathbb{C}}(G) \to \mathrm{Mod}_{\mathbb{C}}(\mathcal{H}(G, K)), \qquad V \mapsto V^K$$

is exact (see, e.g., the introduction of [Sch15]); this provides a useful tool for studying the complex representation theory of a $p$-adic Lie group. By contrast, if one replaces $\mathbb{C}$ by our field $k$ of characteristic $p$, this is deeply false! However, there is a substitution for this fact by [Sch15]: assuming that $K$ is a $p$-torsionfree pro-$p$ group, there is a Hecke differential graded algebra $\mathcal{H}_K^\bullet$ such that there is an equivalence of categories

$$D(G) \xrightarrow{\sim} D(\mathcal{H}_K^\bullet).$$

Here, $D(\mathcal{H}_K^\bullet)$ denotes the derived category of differential graded left $\mathcal{H}_K^\bullet$-modules.

In light of this, it is now a natural question to ask if one can discuss derived variants of the mod $p$ Satake transform introduced by Herzig. The answer is 'yes', and various authors have taken this question in a few different directions, see for example [Ron18] and [KP25]. The most relevant direction for our purposes was taken by Heyer in [Hey23], which we detail below.

3.3. **Notation.** We introduce notation which will hold for the next two subsections. Recall that $k$ is a field of characteristic $p$ and $F/\mathbb{Q}_p$ is a finite extension with integers $\mathcal{O}_F$. Let $\mathbf{G}$ be a connected reductive $F$-group, and set $G = \mathbf{G}(F)$. Fix a parabolic subgroup $P \leq G$ with Levi quotient $M$ and unipotent radical $U$, so that $P = UM$. Let also $K \leq G$ be an open compact modulo center subgroup satisfying the Iwasawa decomposition $G = PK$ and $K_P = K_U K_M$, where as usual we are writing

$$K_P = K \cap P, \qquad K_U = K \cap U, \qquad K_M = K \cap M.$$

Given $V \in \mathrm{Rep}_k(K)$ and $W \in \mathrm{Rep}_k(K_M)$, we define their Hecke algebras to be

$$\mathcal{H}_G(V) \coloneqq \mathrm{End}_G(\mathrm{ind}_K^G V), \qquad \mathcal{H}_M(W) \coloneqq \mathrm{End}_M(\mathrm{ind}_{K_M}^M W).$$

Here, $\mathrm{ind}_{K_M}^M$ denotes the compact induction of a smooth representation of $K_M$. There is also an interpretation of these Hecke algebras as functions similar to the one presented in Subsection 3.1, which we will use later.

An important example of the above setup to keep in mind is where $G = \mathrm{GL}_n(F)$, $P$ is the subgroup of upper triangular matrices, $M = T$ is the subgroup of diagonal matrices, $U$ is the subgroup of upper triangular unipotent matrices, and $K = \mathrm{GL}_n(\mathcal{O}_F)$. Then $K_P$, $K_M$, and $K_U$ are just respectively the upper triangular, diagonal, and upper triangular unipotent subgroups of $\mathrm{GL}_n(\mathcal{O}_F)$. We will later specialize to this case.

3.4. **Heyer's construction of higher mod $p$ Satake transforms.** In [Hey23], Heyer constructs a family of higher mod $p$ Satake transforms as follows. Let $L_U$ denote the left adjoint of the derived inflation functor

$$\mathrm{R}\operatorname{Inf}_M^P : D(M) \to D(P).$$

We then write $L(U,-) \coloneqq L_U \circ \mathrm{R}\operatorname{Res}_P^G : D(G) \to D(M)$ (which is the left adjoint of parabolic induction). Similarly, let $L_{K_U}$ denote the left adjoint of

$$\mathrm{R}\operatorname{Inf}_{K_M}^{K_P} : D(K_M) \to D(K_P),$$

and write $L(K_U,-) \coloneqq L_{K_U} \circ \mathrm{R}\operatorname{Res}_{K_P}^K : D(K) \to D(K_M)$. The proofs of the existence of both of these, as well as lots of other facts about them, are contained in [Hey23].

The key fact which allows Heyer to construct higher Satake transforms is that, for any $V \in D(K)$, there is a natural isomorphism

$$L(U, \operatorname{ind}_K^G V) \cong \operatorname{ind}_{K_M}^M L(K_U, V).$$

Thus, the application of the functor $L(U,-)$ induces a $k$-algebra homomorphism

$$\mathcal{S}_V : \operatorname{End}_{D(G)}(\operatorname{ind}_K^G V) \to \operatorname{End}_{D(M)}(\operatorname{ind}_{K_M}^M L(K_U, V))$$

for any $V \in D(K)$. In particular, if $V \in \operatorname{Rep}_k(G)$, a short argument (essentially provided in [Hey23, Explanation 4.3.3]) shows that taking cohomology in degree $i - \dim(K_U)$ yields a map

$$\mathcal{S}_V^i : \operatorname{End}_{\operatorname{Rep}_k(G)}(\operatorname{ind}_K^G V) \to \operatorname{End}_{\operatorname{Rep}_k(M)}(\operatorname{ind}_{K_M}^M H^i(K_U, V)),$$

which can also be rewritten merely as

$$\mathcal{S}_V^i : \mathcal{H}_G(V) \to \mathcal{H}_M(H^i(K_U, V)).$$

(Here, $H^i(K_U, V)$ denotes the continuous group cohomology of $K_U$ with coefficients in $V$.) These are precisely the higher Satake transforms which we study in [BJPS26].

3.5. **Computing Heyer's Satake transforms.** In [Hey23], Heyer shows that in the extreme degrees of $i = 0$ and $i = \dim(K_U)$, his construction recovers two Satake transforms studied by Herzig, as well as Henniart and Vignéras; he proves this by connecting his derived construction with concrete formulas which coincide with those preciously studied. However, he does not explicitly compute the interior degrees, where novel Satake transforms lie.

We now proceed with explaining our contributions. Everything in this section is joint work with Hargun Bhatia, Yashi Jain, and Ben Savoie, and is contained in our preprint [BJPS26]. Our goal for this paper was to compute the interior higher Satake transforms in as much detail as possible, and try to answer natural questions one would have about them.

We first obtain two formulas for the higher Satake transforms. One is similar to the formulas proven in [Hey23] for the Satake transforms in extreme degrees, and expresses $\mathcal{S}_V^i$ as a sum over double-cosets via well-known cohomological operations:

**Theorem 3.1** (BJPS)**.** *Let $V \in \operatorname{Rep}_k(K)$. For $\Phi \in \mathcal{H}_G(V)$ and $\xi \in H^i(K_U, V)$, we have*

$$\mathcal{S}_V^i(\Phi)([1,\xi]) = \sum_{g \in K_P \backslash P / K_U} \left[ \mathrm{pr}_M(g^{-1}), \operatorname{cores}_{g^{-1}}^{(i)} \left( \mathrm{pr}_{g^{-1}}^{(i)}(\Phi([1,\xi])) \right) \right].$$

*Here $\mathrm{pr}_{g^{-1}}^{(i)}$ is a projection arising from a certain Mackey decomposition, and $\operatorname{cores}_{g^{-1}}^{(i)}$ is a corestriction map with target $H^i(K_U, V)$.*

While this result is pleasing as it generalizes known results, we found it desirable to have explicit formulas at the level of cochains in order to perform concrete computations. For our second formula, we define (on the level of cochains) an explicit map

$$C_m^{(i)} : H^i\left(K_U, \operatorname{ind}_{K_P}^P V\right) \longrightarrow H^i(K_U, V)$$

for any $m \in M$, and we prove the following, using now the function model of the Hecke algebras:

**Theorem 3.2** (BJPS)**.** *Let $V \in \mathrm{Rep}_k(K)$. For $\Psi \in \mathcal{H}_G(V)$, viewed now as a function $G \to \mathrm{End}_k(V)$, and given $\xi \in H^i(K_U, V)$, we have*

$$\mathcal{S}_V^i(\Psi)(m)(\xi) = C_m^{(i)}\left(\Psi(-)(\xi)\right).$$

Next, we specialize to the case of $G = \mathrm{GL}_n(F)$. For $1 \leq j \leq n$, let $T_j$ be the standard Hecke operator supported on $Kt_jK$, where

$$t_j \coloneqq \mathrm{diag}(\underbrace{\varpi, \ldots, \varpi}_{j}, 1, \ldots, 1).$$

If $I \subseteq \{1, \ldots, n\}$ has cardinality $j$, let $t_I$ be the diagonal matrix with $\varpi$ in the positions indexed by $I$ and 1 elsewhere, and set $K_{U,I} \coloneqq t_I K_U t_I^{-1} \cap K_U$. Our maps $C_m^{(i)}$ of Theorem 3.2 are easier to compute in this particular case on the standard Hecke operators, and we are able to prove the following:

**Theorem 3.3** (BJPS)**.** *For every $0 \leq i \leq \dim(K_U)$ and every $1 \leq j \leq n$,*

$$\mathrm{Supp}\, \mathcal{S}_V^i(T_j) \subseteq \bigsqcup_{\substack{I \subseteq \{1,\ldots,n\} \\ \#I = j}} K_T t_I.$$

*For each subset $I$ of cardinality $j$,*

$$\mathcal{S}_V^i(T_j)(t_I) = \mathrm{cores}_{K_{U,I}}^{K_U} \circ T_{j,I},$$

*where $T_{j,I} : H^i(K_U, V) \to H^i(K_{U,I}, V)$ is induced by conjugation by $t_I^{-1}$ and the coefficient map $T_j(t_I)$.*

When $n = 2$, i.e., for $G = \mathrm{GL}_2(F)$, the cohomology of $K_U$ is much easier to work with than for $n > 2$, and we are able to perform very detailed computations with the expressions appearing in Theorem 3.3. Write $[F : \mathbb{Q}_p] = ef$, where $e$ and $f$ are the ramification index and residue degree, respectively. For every Serre weight $V$, the corresponding Hecke algebra of $G$ is isomorphic to

$$\mathcal{H}_G(V) \cong k[T_1, T_2^{\pm 1}].$$

For a sufficiently regular Serre weight (i.e., one which is non-Steinberg), it turns out that $T_2$ remains invertible under every higher Satake transform, and the image of $T_1$ becomes nilpotent in every interior degree, in stark contrast to the extreme degrees (where the Satake transforms are injective).

**Theorem 3.4** (BJPS)**.** *Let $V$ be a non-Steinberg Serre weight. For every $1 \leq i \leq ef - 1$, there is a positive integer $R_i(V)$ such that there is an isomorphism*

$$\mathrm{im}\, \mathcal{S}_V^i \cong k[x, y^{\pm 1}]/\left(x^{R_i(V)}\right), \qquad \mathcal{S}_V^i(T_1) \mapsto x, \quad \mathcal{S}_V^i(T_2) \mapsto y.$$

*The integers $R_i(V)$ satisfy $R_i(V) = R_{ef-i}(V)$ and*

$$R_i(V) \in \{e - 1 - r_i,\ e - r_i,\ e + 1 - r_i\} \qquad \textit{where} \qquad r_i \coloneqq \left\lfloor \frac{\min\{i, ef - i\} - 1}{f} \right\rfloor.$$

*Moreover, these bounds are sharp: there are examples obtaining all three possible values of $R_i(V)$.*

Finally, if $F/\mathbb{Q}_p$ is in fact unramified, we are almost always able to determine the integers $R_i(V)$.

**Corollary 3.5** (BJPS)**.** *Assume that $F/\mathbb{Q}_p$ is unramified, so $e = 1$. Then for every $1 \leq i \leq f - 1$, we know $R_i(V) \in \{1, 2\}$. For sufficiently general Serre weights, in fact $R_i(V) = 1$ for all such $i$.*

3.6. **Future directions.** The most immediate future direction to take this research would be to perform computations similar to Theorem 3.4 for $\mathrm{GL}_n$ where now $n > 2$. In [BJPS26], we show for $\mathrm{GL}_3(F)$ and $k$ the trivial one-dimensional representation of $\mathrm{GL}_3(\mathcal{O}_F)$ that $\mathcal{S}_k^1(T_1)$ is not nilpotent, demonstrating that the nilpotence phenomenon described in Theorem 3.4 does not generalize in any particularly obvious way to $\mathrm{GL}_n$ for $n > 2$. Thus, the behavior of these higher Satake transforms in larger reductive groups has the potential to be very different than for $\mathrm{GL}_2$. One computation which has the potential to be both enlightening and highly feasible would be to compute the image of $S_V^i$ modulo nilpotents for each $i$ and almost all $V$ for $\mathrm{GL}_n(F)$.

Another potential direction of research which is more combinatorial would be to investigate the integers $R_i(V)$ appearing in Theorem 3.4 in more detail. There are three values which they take on; can we classify precisely when they are each potential value given a particular Serre weight $V$? As stated in that result, we have examples which exhibit all three values, and it is very easy to test for more Serre weights what the results would be.

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